

Worksheet · Section 4.2–4.3

Fault Tolerance & Parallel Computing



The Big Picture

Good systems keep working when parts fail, and big problems get faster when work is split up. Practice reading network diagrams and calculating parallel run times and speedup.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|---|---|
| _____ 1. Redundancy | A. Extra time spent coordinating processors or devices |
| _____ 2. Fault tolerance | B. Operations run one at a time, in order |
| _____ 3. Single point of failure | C. The ability to keep working when some parts fail |
| _____ 4. Path | D. One part whose failure stops or disconnects the system |
| _____ 5. Sequential computing | E. A program runs across multiple devices |
| _____ 6. Parallel computing | F. Sequential time divided by parallel time |
| _____ 7. Distributed computing | G. Extra components or paths used as backups |
| _____ 8. Speedup | H. A sequence of connected devices from sender to receiver |
| _____ 9. Processor | I. The part of a computer that carries out instructions |
| _____ 10. Overhead | J. Some operations run at the same time on multiple processors |

Part 2 · Fill It In

Use each word from the word bank once.

sum

redundancy

speedup

reroute

fault-tolerant

cost

longest

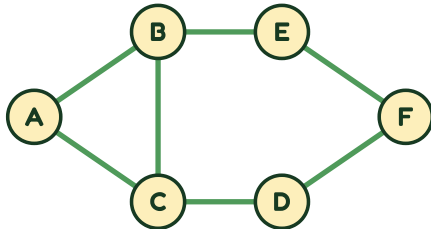
sequential

- Extra components used as backups are _____.
- A system that keeps working when parts fail is _____.
- When a connection fails, the Internet can _____ data along another path.
- Redundancy increases reliability but also increases _____.
- A sequential solution takes the _____ of all its task times.
- A parallel solution takes as long as its _____ parallel task (plus any sequential parts).
- Sequential time ÷ parallel time = _____.
- The _____ portion of a program limits how much parallel computing can help.

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Practice

Part 3 · Network Analysis



- How many different paths go from A to F (no device repeated)? _____
- Can any ONE connection failure disconnect A from F? _____
- Minimum number of connections that must fail to disconnect A from F: _____ (for example, _____)
- Is there a single device (other than A or F) whose failure disconnects them? _____

Part 4 · Speedup Math

Four independent tasks take 40, 30, 20, and 10 seconds.

Setup	Time
One processor	_____
Two processors (best split)	_____ (_____ and _____)
Speedup with two	_____
Four processors	_____
Speedup with four	_____

Part 5 · Dependencies

Task A (15 s) must finish before tasks B (25 s) and C (35 s) can start. B and C are independent. Task D (10 s) needs results from both B and C.

Sequential time: _____ s

Two processors: _____ + _____ + _____ = _____ s

Speedup: about _____

Would a third processor help?

Part 6 · Design for Failure

A school's network connects every classroom to the Internet through one router. Identify the problem, propose a fix, and describe one trade-off.



Think:

Parallel time = sequential parts + the LONGEST parallel part. Adding processors can't make a program faster than its sequential portion.

Name: _____

Class: _____

Date: _____

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What is redundancy?
- A. Including extra components that can be used if others fail
 - B. Sending data twice as fast
 - C. A type of compression
 - D. Removing extra devices

★★ On Level

2. Three independent tasks take 12, 18, and 9 seconds. How long do they take on three processors?
- A. 12 s
 - B. 18 s
 - C. 39 s
 - D. 9 s

★★ On Level

3. Which is an example of distributed computing?
- A. One laptop running a spreadsheet
 - B. A single phone game
 - C. A research project that uses thousands of volunteers' computers
 - D. A calculator app

★★★ Exam Hard

4. A program has 40 s of sequential work and 80 s of work that can be split evenly. How long does it take on 4 processors?
- A. 120 s
 - B. 40 s
 - C. 30 s
 - D. 60 s

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)
- A. Multiple paths between devices make a network more fault-tolerant.
 - B. Parallel solutions are always slower than sequential ones.
 - C. Adding processors can't reduce time below the sequential portion.
 - D. Redundancy removes all possible failures.

Part 8 · Written Response

Explain why the speedup of a parallel solution eventually levels off as you add more processors. Use numbers from Part 4 or Part 5.
